PolyPLASTY PolyPOM Polyoxymethylene (POM)

Categories: Polymer; Thermoplastic; Acetal (Polyoxymethylene, POM); Acetal Copolymer, Unreinforced

Material Notes: POM has minimal water absorption, excellent dimensional stability and the best properties for chip machining. Thanks to its favorable characteristics, great hardness, rigidity and strength with good toughness, chemical resistance and its shear behavior with abrasion resistance, POM is a successful replacement for metal materials. It is resistant to organic solvents. It also features good resistance to black light UV-radiation.

Information provided by PolyPLASTY.

Vendors: No vendors are listed for this material. Please click here if you are a supplier and would like information on how to add your listing to this material.

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Physical Properties	Metric	English	Comments
Water Absorption	0.20 %	0.20 %	under regular conditions; ISO 62
Moisture Absorption at Equilibrium	0.80 %	0.80 %	ISO 62

Mechanical Properties	Metric	English	Comments
Hardness, H358/30	145 MPa	21000 psi	ISO 2039-1
Hardness, Shore D	81	81	ISO 868, ISO 2039-2
Elongation at Break	27 %	27 %	ISO 527
Elongation at Yield	9.4 %	9.4 %	ISO 527
Modulus of Elasticity	2.70 GPa	392 ksi	ISO 527
Flexural Strength	97.0 MPa	14100 psi	ISO 178
Fatigue Strength	31.0 MPa	4500 psi	flexure; ASTM D671
Izod Impact, Notched	0.600 J/cm	1.12 ft-lb/in	ISO 180/4A
Izod Impact, Notched (ISO)	5.50 kJ/m² @Temperature 23.0 °C	2.62 ft-lb/in² @Temperature 73.4 °F	ISO 180/1A
	5.50 kJ/m² @Temperature -30.0 °C	2.62 ft-lb/in² @Temperature -22.0 °F	ISO 180/1A
Izod Impact, Unnotched (ISO)	100 kJ/m² @Temperature -30.0 °C	47.6 ft-lb/in² @Temperature -22.0 °F	ISO 180/1C
	930 kJ/m² @Temperature 23.0 °C	443 ft-lb/in² @Temperature 73.4 °F	ISO 180/1C
Charpy Impact Unnotched	19.0 J/cm² @Temperature -30.0 °C	90.4 ft-lb/in² @Temperature -22.0 °F	ISO 179/1eU
	21.0 J/cm² @Temperature 23.0 °C	99.9 ft-lb/in² @Temperature 73.4 °F	ISO 179/1eU
Charpy Impact, Notched	0.550 J/cm² @Temperature -30.0 °C	2.62 ft-lb/in² @Temperature -22.0 °F	ISO 179/1eA
	0.600 J/cm² @Temperature 23.0 °C	2.86 ft-lb/in² @Temperature 73.4 °F	ISO 179/1eA
Tensile Creep Modulus, 1000 hours	1400 MPa	203000 psi	0.5 percent; ISO 899-1

Electrical Properties	Metric	English	Comments
Electrical Resistivity	1.00e+14 ohm-cm	1.00e+14 ohm-cm	IEC 60093
Surface Resistance	1.00e+14 ohm	1.00e+14 ohm	IEC 60093
Dielectric Constant	3.8 @Frequency 50 Hz	3.8 @Frequency 50 Hz	IEC 60250
	3.8 @Frequency 1e+6 Hz	3.8 @Frequency 1e+6 Hz	IEC 60250
Dielectric Strength	40.0 kV/mm @Thickness 1.00 mm	1020 kV/in @Thickness 0.0394 in	IEC 60243-1
Dissipation Factor	0.0010 @Frequency 50 Hz	0.0010 @Frequency 50 Hz	IEC 60250
	0.0050 @Frequency 1e+6 Hz	0.0050 @Frequency 1e+6 Hz	IEC 60250
Comparative Tracking Index	600 V	600 V	IEC 60112

Thermal Properties	Metric	English	Comments
CTE, linear, Transverse to Flow	110 µm/m-°C	61.1 µin/in-°F	ISO 11359
Specific Heat Capacity	1.47 J/g-°C	0.351 BTU/lb-°F	IEC 1006
Thermal Conductivity	0.310 W/m-K	2.15 BTU-in/hr-ft²-°F	DIN 52 612
Maximum Service Temperature, Air	90.5 °C	195 °F	long term
	140 °C	284 °F	short term
Vicat Softening Point	150 °C @Load 5.10 kg	302 °F @Load 11.2 lb	point B; ISO 306 VST/B/50
Minimum Service Temperature, Air	-50.0 °C	-58.0 °F	
Flammability, UL94	HB @Thickness 0.800 mm	HB @Thickness 0.0315 in	IEC 60695-11-10

Descriptive Properties			
LOI Acid Index (%)		15	ISO 4589

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